# Normalization of Stain and Scan Distortions for Red Blood Cell Analysis on Manually Prepared Smears Using Deep Learning

Mohamed Elmanna [a], Ahmed Elsafty [b], Yomna Ahmed [b], Muhammad Rushdi [a, c*], and Ahmed Morsy [a]

[a] *Department of Biomedical Engineering and Systems, Faculty of Engineering, Cairo University, Cairo University St, Giza 12613, Egypt*

[b] *PathOlOgics, LLC for processed medical data and tele-pathology services, Egypt*

[c] *School of Information Technology, New Giza University, Giza 12256, Egypt*

**Abstract**

*Background:* The introduction of the digital pathology realm has provided numerous opportunities in computer-aided diagnosis (CAD) and many tools that can assist pathologists. Such valuable tools include the utilization of advances in machine learning techniques, including deep learning (DL). In hematopathology, however, staining variations and distortions caused by digital pathology scanners during image digitization pose challenges to DL-based automated digital pathology systems. These challenges are the most likely cause for CAD performance degradation.

*Objectives:* In this work, we investigate the effects of stain and scan normalization on red blood cell (RBC) image analysis systems. The normalized images are thus used for DL-based RBC segmentation and classification.

*Methods:* A highly diverse dataset of more than 100K RBCs named 100K-RBC-PathOlOgics was used for RBC segmentation and classification. Then, large simulated/synthetic datasets were created to reflect the wide data variability among scanners, staining protocols, and imaging conditions. Subsequently, an autoencoder with a DL architecture was trained using the simulated data to perform normalization of stain and scan distortions. The cell segmentation and classification stages were realized using U-Net and EfficientNetB0 architectures, respectively. The proposed scheme was validated on real and simulated datasets from four different scanners to assess the impact of autoencoder-based normalization on RBC classifier performance.

*Results:* The results show that our DL-based normalizer can significantly improve the RBC classification performance (in terms of the F1-score) against the RBC classifier without image normalization. The improvements are especially significant on test images collected from the two scanners with the low-quality images, namely, Scanner 3 (with a *p*-value of $\mathbf{2.0 \times 10^{-2}}$) and Scanner 4 (with a *p*-value of $\mathbf{4.1 \times 10^{-8}}$).

*Conclusions:* The experimental results highlight the importance of RBC image normalization in building robust and generalizable DL-based models for RBC classification. Moreover, our DL-based normalization approach outperforms two state-of-the-art methods on real and simulated datasets for RBC classification.

***Keywords:*** Whole slide images, hematopathology, stain normalization, scanner distortion normalization, deep learning, RBC classification.

## I. INTRODUCTION

In many countries, the complete blood count (CBC) test is among the top laboratory tests by volume [1]. In the United States, for example, up to 17.3% of performed CBC tests require morphological examination of blood cell images under a microscope or using automated scanners [2]. The use of digital scanners, which introduced the realm of digital pathology, has created a profound opportunity for the use of machine learning and deep learning (ML/DL) technologies [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. Currently, blood-related cancer cases represent about 6.5% of the total cancer cases reported worldwide [12]. Once their success is proven, ML/DL technologies can create substantial opportunities for reducing pressure on pathologists, reducing human errors, and improving the throughput and profitability of pathology labs, which are already suffering from shortage of medical laboratory professionals. A major limitation associated with the quality in hematopathology comes from the critical preanalytical staining step in the sample processing chain. Once a blood droplet (10-50uL) is placed on a glass slide and smeared out to produce a thin layer of blood, a dye is applied to induce color variations to characterize different intracellular structures. Good staining enables pathologists to differentiate these structures well, relying on different degrees of red, blue, violet, gray, and orange colors. As this staining step is neither easy to perform nor standardized, it typically results in substantial image variability that complicates the job of pathologists and ML-based digital pathology systems [13]. While image digitization using optical scanning introduces additional image variability, distortions introduced by such scanners are progressive by usage time and fairly consistent with the same scanner. The same may occur with the automated staining systems which have been developed to reduce manual variabilities. However, automated staining systems are still not widely deployed, particularly in lower-income countries. Normalizing variabilities caused by staining and scanning may prove to be a critical success factor for the implementation of any ML/DL system applied to whole slide images (WSIs). Theoretically, without such normalization, there is no performance generalizability of any AI-DL model for object detection and image classification. For instance, despite using more than 1,000 WSIs (1,178,408 annotated cells after augmentation) but with a lack of stain and scan normalization, an end-to-end ML/DL-based approach for automated morphology did not achieve more than 0.75 mAP and 0.78 F1-score for cell detection and classification, respectively [8]. Such performance suffers from limited daily work applicability and requires substantial manual review of cases analyzed by such models for fear of missed findings or passed false findings. In this work, we show that normalization of stain and scan distortions can significantly improve the performance of a DL-based system designed for RBCs classification.

## II. Related Work

Deep learning (DL) models have demonstrated a significant promise in tackling RBC analysis problems such as cell detection and counting [14], [15], [16], cell segmentation [17], [18], [19], and cell categorization [20], [21], [22], [23]. These models are trained and tested under the assumption that training and testing sets come from independently and identically distributed data (IID). However, the IID assumption may not hold in many real-world scenarios. Domain generalization (DG) aims to train robust models by using data from different domains to generalize to new unseen data. The surveys by Zhou *et al.* [24] and Wang *et al.* [25] provide an in-depth literature review on this topic.

Automated systems for RBCs classification may be trained on datasets originating in different labs using different setups. This results in variability in image characteristics in such heterogeneous datasets, which can negatively affect the final classification results. A normalization step is thus considered a necessary and critical preprocessing stage.

Zhang *et al.* [26] investigated the corruption types in peripheral blood smear images and established a simulated benchmark dataset called Smear-C. The authors proposed an AugMix-inspired augmentation policy and a special transformation to resist smear corruptions. The augmentation policy mixes several chains of augmentation operations and adds the resulting mix to the original image. A total of 16,013 white blood cell (WBC) images were used to evaluate resistance to 16 types of smear corruption. However, this study focused only on WBC images and carried out performance evaluation on simulated data only with no visual review by hematopathologists.

All publicly available RBC datasets are acquired by one scanner/camera, which is insufficient to address domain generalization in RBC classification. Because of the limitations in public RBC datasets in terms of size and diversity, we previously introduced a new large, multi-scanner RBC dataset named 100K-RBC-pathOlOgics [27]. To the best of our knowledge, this is the first effort to address domain generalization in RBC classification.

The purpose of this paper is to study the effect of stain and scan normalization on RBC classification performance. The RBC image analysis system consists of three consecutive stages for stain and scan distortion normalization, RBC segmentation, and RBC classification. Each of these stages was realized through a carefully selected DL architecture. In addition to the previously introduced large and diverse dataset [27], simulated/synthetic datasets of up to 971K images were used to cover wide variations in the intracellular and background color, contrast, brightness, and saturation to account for the substantial image variabilities between different scanners and staining protocols. The major contributions of this paper are two folds. First, the paper introduces extensive experiments to study the effect of normalization of stain and scan distortions on performance of RBC classification using real and simulated images. Second, simulated/synthetic datasets of up to 971K synthetic images were developed by considering several factors to train DL-based normalization models that boost the RBC classification performance in multi-source domain generalization scenario.

## III. METHODOLOGY

To show the effect of staining and scanning normalization on the performance of RBC classification, we employed the system depicted in Fig. 1(c). The system employed three DL architectures to accomplish its tasks. A convolutional autoencoder model was utilized to perform stain normalization and scanning distortion correction. RBC segmentation was carried out using the U-Net architecture, while the EfficientNetB0 model was utilized for RBC classification. The system was trained using images from three different scanners and tested using a fourth scanner. We conducted a comparison between the performance of RBC classification with and without the autoencoder-based stain and scan normalization. In the following section, we will introduce the used datasets.

### *A. Real multi-scanner RBC image datasets*

The authors collected peripheral-blood and bone-marrow WSIs of different 25 patients from four digital pathology scanners to create two datasets for RBCs segmentation and classification.

The segmentation dataset, named "100K-RBC-Mask-PathOlOgics", has 100,118 cropped RBC images with their corresponding masks. The second dataset, named "100K-RBC-PathOlOgics", has 100,873 RBC images and was utilized for image classification. Each cell was categorized into one of eight classes (normal RBCs, ovalocytes, burr cells, fragmented RBCs, teardrops, two overlapped RBCs, three overlapped RBCs and other RBC types). There is also an extended dataset of 76,104 RBC images (whose characteristics are similar to the 100K-RBC_PathOlOgics dataset) used only for classification performance testing. This test dataset was acquired from the same patients using the same four scanners. Four types of distortion (namely, color, overexposure, underexposure, and defocus) were also simulated to validate the classification model performance on both real and simulated data.

The RBC images were collected with four different digital pathology scanners using four different light microscopes with 40X magnification power, control units, integrated cameras on top, and software for stitching the patches altogether. Each one of the acquired 25 WSIs contained the eight RBC classes with variable extents. Therefore, images for each RBC class were collected from each WSI. We denoted the data subsets collected from the four scanners as real dataset 1 (RDS1), real dataset 2 (RDS2), real dataset 3 (RDS3), and real dataset 4 (RDS4), respectively. The RDS3 images show the widest staining and dyeing variability, while the RDS4 images show the highest blurring variability. More information about these datasets is reported in Appendix-A.

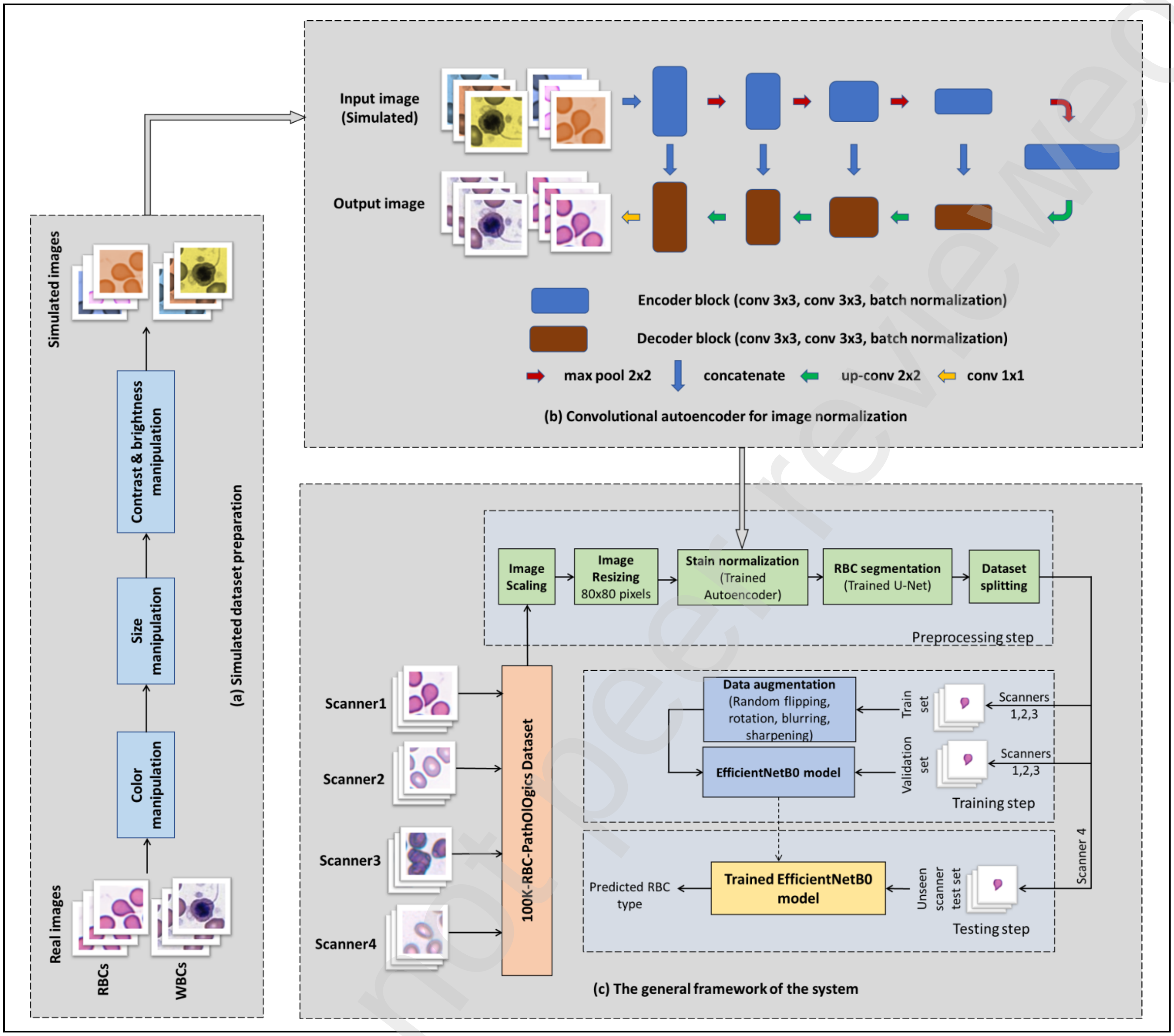



Figure 1. The general framework for the proposed RBCs classification system with a deep-learning-based image normalization module, (a) Simulated/synthetic dataset preparation, (b) Convolutional autoencoder for image normalization, (c) The general framework of the proposed system.

### B. *Simulated/synthetic blood cell image datasets*

Because it is not feasible to collect real images with full distortion spectrum of each cell, the authors suggested and used an equation (Eq. 1) to reverse the process and synthesize a large number of variations by manipulating the best quality target images. A DL-autoencoder normalizer was then trained to restore the target images from the synthetically distorted/corrupted images. The idea was to manipulate the target images by overlaying the major 12 colors with 27 levels of brightness, contrast, and saturation to generate intracellular and/or background distortions (See Eq. 1 and Fig. 1 (a)).

We developed nine large simulated/synthetic datasets during a series of experiments to assess several factors in the data. Examples of these variations are shown in Fig. 2. Each of the nine datasets shows different target images selection and hence normalization variation (See Table II).

The image manipulation process can be mathematically described as follows (Eq. 1). Given a corruption-free $m \times n$ image $I$, a corresponding corrupted image, $\acute{I}$, is defined as follows,

$$\acute{I}(i,j) = \alpha_1 I(i,j) + \alpha_2 C, \quad (1)$$

where $\alpha_1$ is an image intensity scaling factor in the range [0.5, 1.3], $\alpha_2$ is a color scaling factor in the range [0.1, 0.6], and $C$ is an overlaid color image. The RGB color values are varied within the following set $C = (C_{blue}, C_{green}, C_{red}) \epsilon$ $\{$(255,0,0), (0,255,0), (0,0,255), (0,255,255), (255,0,255), (255,255,0), (0,255,100), (0,100,255), (100,255,0), (255,100,0), (255,0,100), (100,0,255)$\}$.

Pairs of $(\alpha_1,\alpha_2)$ values determine the manipulation severity level. For example, the pair $(\alpha_1,\alpha_2) = (0.5,0.5)$ leads to the most severe background discoloration, $(\alpha_1,\alpha_2) = (1.3,0.3)$ produces the most severe intracellular discoloration, while $(\alpha_1,\alpha_2) = (1.0,0.1)$ causes relatively low manipulation (See Fig. 3). The pairs of $(\alpha_1,\alpha_2)$ values are varied within the following set: $(\alpha_1,\alpha_2)$ ϵ$\{$(1.3,0.3), (1.25,0.3), (1.2,0.3), (1.15,0.25), (1.2,0.25), (1.25,0.25), (1.1,0.1), (1.1,0.15), (1.1,0.2), (1.15,0.1), (1.15,0.15), (1.15,0.2), (1,0.1), (1,0.15), (0.9,0.1), (0.8,0.2), (0.7,0.3), (0.6,0.4), (0.5,0.5), (0.9,0.2), (0.9,0.3), (0.8,0.3), (0.8,0.4), (0.7,0.4), (0.7,0.5), (0.6,0.5), (0.6,0.6)$\}$.

In terms of target image selection, we had to synthesize a large number of distortions for each target image leading to a limited number of target images. the selection of such images was found to have a critical influence on the normalization outcomes. Thus, the authors passed through nine complete multi-directional interactive experiments and stringent root cause analysis to reach the best factors for selecting the target images.

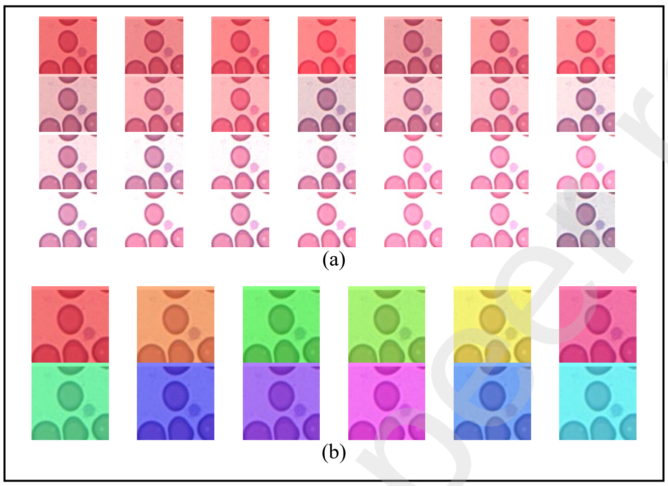



Figure 2. Simulated/synthetic blood cell images. (a) using overlaid red color and different manipulation severities, (b) using different overlaid color variations and fixed manipulation severity $(\alpha_1,\alpha_2) = (0.5,0.5)$. the lower right image in (a) is the original image.

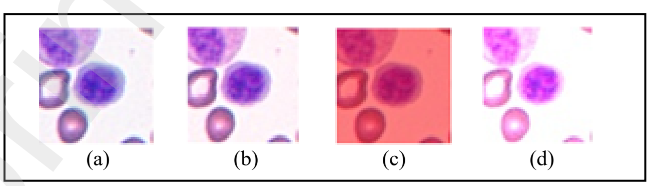



Figure 3. Simulated/synthetic blood cell images with different manipulation factors: (a) Original image, (b) The image in (a) after relatively low color manipulation, (c) The image in (a) after severe background discoloration, (d) The image in (a) after severe intracellular discoloration.

As a result, nine different normalization modules and their corresponding simulated/synthetic datasets were developed and tested one by one but under different factors (listed in Table II). The basic influencing factors were discovered to be related to the number of selected RBCs to WBCs ratio (RBCs-to-WBCs ratio), magnification levels, images from different WSIs and scanners, the number of cells in the image, and the proportions of the eight cell types in the created data. Table II describes the nine simulated/synthetic datasets.

### *C. Effect of scanner variability on RBC classification*

As shown in Fig. A.4, the RBC images obtained from each of the four different scanners show distinct patterns. These patterns can affect the performance of RBC classifiers. Indeed, if images for one scanner were not included in the training data of a classifier, this classifier would have degraded generalization performance on the images of that scanner.

We studied the domain shift effect of training a model using images from multi-scanners and testing using images from unseen scanner data on the classification performance.

### *D. Stain and scan normalization training using simulated/synthetic data*

We seek to build a DL model for blood cell image normalization. This model should serve as a preprocessing module before image segmentation and classification. This module was designed to eliminate the staining/dyeing variabilities, to improve the performance generalizability especially on images from unseen digital scanners. It was also created to boost the performance on images with high variabilities in color, contrast, brightness, and/or saturation.

The proposed deep-learning (DL) normalizer was constructed based on a convolutional autoencoder architecture with skip connections which showed promising results in applications of image quality restoration [28]. A convolutional autoencoder network is a type of CNN that consists of two stages: an encoding stage where the input image is converted into a feature representation or a compressed version of the input, and a decoding stage to reconstruct or restore the input from the compressed input representation. The autoencoder architecture is composed of 4 blocks in the encoding stage, where each block has two convolutional layers and one max-pooling layer. For the decoding stage, there are 4 blocks, where each one is composed of a transposed convolutional layer for feature map upsampling, concatenated with the output of the parallel block in the encoder and followed by two convolutional layers. The model has input and output fixed sizes of 80×80×3. Also, the numbers of feature channels used in the convolutional layers are 32, 64, 128 and 256, respectively, for the encoder and decoder blocks. The overall architecture of the proposed normalizer was shown in Fig. 1(b).

For the encoding stage, an input image $I = \{I_r, I_g, I_b\}$ and $n$ convolution filters $f^{(e)} = \{f_1^{(e)}, f_2^{(e)},\ldots,f_n^{(e)}\}$ were used to generate the intermediate feature maps

$$F_m = s\left(I * f_m^{(e)} + b_m^{(e)}\right), m = 1, 2, 3,\ldots,n$$

where s is the activation function and $b_m^{(e)}$ denotes the bias of the $m^{th}$ filter at the encoder side.

For the decoding stage, a normalized image was obtained by the convolution of the intermediate feature maps $F_m$ with decoder-side convolution filters $f_m^{(d)}$ ,$m = 1,2,\ldots,n$:

$$\tilde{I} = s\left(F_m * f_m^{(d)} + b_m^{(d)}\right)\ m = 1, 2, 3,\ldots,n$$

where $b_m^{(d)}$ denotes the bias of the $m^{th}$ filter at the decoder side.

The mean-square error (MSE) was used as the loss function $\ell(\tilde{I},I)$ between the normalized images $\tilde{I}$ and the corresponding ground-truth images $I$. Given a corruption-free $c \times c$ image $I$ and its restored approximation $\tilde{I}$, the MSE between the two images is defined mathematically as

$$\ell(\tilde{I},I) = \frac{1}{c \times c}\sum_{i=0}^{c-1}\sum_{j=0}^{c-1}\left[I(i,j) - \tilde{I}(i,j)\right]^2 . \qquad (2)$$

In our work, the autoencoder was independently trained by each of the simulated/synthetic image datasets (listed in Table II), covering wide variability ranges. These normalization models were trained by both RBC and WBC images to ensure no discoloration between them. During the training of the stain and scan normalization model, different levels of blurring and sharpening were added randomly through augmentation steps. Qualitative model evaluation was performed to reliably check the samples for artifacts, e.g., ensuring that no confusion/false discoloration of the platelets (small blue cells) or the fragmented RBCs (small red cells) were present. Furthermore, normalization models were tested on real images as well as on other simulated images with different staining/dyeing variations. Nine different normalization models were built, one for each of the nine simulated/synthetic datasets in Table II. Each of these normalization models was subsequently used to normalize the real RBC images, and create a corresponding deep–learning–based model for RBC image segmentation and classification.

### *E. Semantic segmentation and classification of RBCs*

We utilized the U-Net architecture [29] for the RBC image segmentation step, which is considered one of the well-known approaches for medical image segmentation. We used the same architecture design in our previous work [27]. The dataset used for the U-Net training was the 100K-RBC-Mask-PathOlOgics (Section III.A). This dataset was split into independent training, validation, and testing subsets with 80%, 10%, and 10% of the data, respectively.

We employed transfer learning with the pre-trained EfficientNetB0 network [30], which was initially trained on the ImageNet dataset, to classify RBC images. Then, all layers were unfrozen to allow fine-tuning of all network layers. This model showed high competence in both classification performance and computational cost when compared against eight state-of-the-art CNN models in our previous study [27]. EfficientNet is a family of CNN architectures built employing a set of scaling coefficients that enable consistent scaling of network depth, width, and resolution. EfficientNetB0, the fundamental structure of the EfficientNet family, in particular, was developed utilizing a multi-objective neural architecture search that optimizes accuracy and FLOP count [30].

The 100K-RBC-pathOlOgics dataset was used for classifier training with eight RBC classes. Guided by results from Table I, we selected the datasets with low classification performance for testing subsets. Either RDS3 or RDS4 is used as the test set, while the remaining images are split using a $5\times2$ cross-validation into independent training and validation subsets.

### *F. Implementation*

The Keras and TensorFlow libraries were used to implement all DL models. For model optimization, the Adam optimizer was utilized. The autoencoder's learning rate (LR) was set to 1e-4, while the segmentation and classification models' LRs were set to 4e-4. If the loss did not improve by at least 1e-4 after each four epochs, the LR was dropped by a factor of ten. If no loss reduction happened for 10 epochs, the training was terminated. All input images were scaled between 0 and 1 and a batch size of 32 was used.

## IV. RESULTS

### *A. Effect of scanner variability on RBC classification*

As presented in Section III.C, we studied the effect of scanner variability on RBC classification performance. Four classification models were constructed. Each model was trained on images collected from only three scanners and tested on images obtained from the fourth scanner. For comparison, a general model satisfying the IID assumption was also trained and tested on images obtained from all four scanners.

The classification results presented in Table I show major differences in the testing accuracy between the four models. On the one hand, the wide staining/dyeing variability in RDS3 led to the lowest classification accuracy (86.47%) for Model 3. On the other hand, the blurring variability in RDS4 led to a moderate testing accuracy (93.65%) for Model 4. Models 1 and 2 showed robust performance (compared to Models 3 and 4) with testing accuracies of 96.6% and 96.13%, respectively.

TABLE I. GENERALIZATION PERFORMANCE OF RBC IMAGES CLASSIFIERS ON SCANNER IMAGES NOT REPRESENTED IN THE TRAINING DATA.

| Model | Model 1 | Model 2 | Model 3 | Model 4 | General Model |
|---|---|---|---|---|---|
| Training set | RDS2, 3, 4 | RDS1, 3, 4 | RDS1, 2, 4 | RDS1, 2, 3 | RDS1, 2, 3, 4 |
| Test set | RDS1 | RDS2 | RDS3 | RDS4 | RDS1, 2, 3, 4 |
| Training accuracy | 98.15% | 97.84% | 98.23% | 98.11% | 98.03% |
| Validation accuracy | 96.72% | 96.40% | 97.02% | 96.75% | 96.24% |
| Test accuracy | 96.60% | 96.13% | 86.47% | 93.65% | 96.07% |

### B. *Evaluation of DL-based stain and scan normalization models*

Nine simulated/synthetic datasets were used to train nine stain and scan normalization models. In Table II, we describe all simulated/synthetic datasets used for building the normalization models with convolutional autoencoders. The evaluation was performed qualitatively through visual examination and measured quantitatively based on the improvement of the RBCs classification accuracy.

Samples of RBCs, WBCs, platelets, and artifacts were reviewed and analyzed to ensure that there were no discoloration patterns among them. For example, a fragmented RBC discoloration pattern can lead to confusion with the platelets, while discolored artifacts can be misclassified as fragmented RBCs or platelets. Evaluation results of the developed stain and scan normalization models are presented in Table II. For each model (Column 1), we list the used dataset (Column 2), dataset description (Column 3), training image characteristics (Columns 4-12), test accuracies on the RDS3 and RDS4 datasets (Columns 13-14), and visual examination remarks (Column 15). Next, we discuss in detail the training setup and results for each of the normalization models.

TABLE II. DESCRIPTION OF NINE SIMULATED/SYNTHETIC DATASETS OF BLOOD CELL IMAGES, THE TRAININIG SETUP, AND THE EVALUATION RESULTS OF NINE STAIN AND SCAN NORMALIZATION MODELS. THE BOLD VALUES INDICATE THE LARGEST VALUES IN THE COLUMN.

| Normalization Model | Normalization Dataset | Description | No. of Target RBCs | No. of Target WBCs | RBC- WBC Ratio | Dataset Size | Scanner Type | Non-RBC-100K WSI training data | Different Zoom | Image Resizing | Blurring and sharpening augmentation | Classification Accuracy (RDS3) | Classification Accuracy (RDS4) | Visual Examination Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| No normalization | NA | | | | | | | | | | | 86.97±0 .0.90% | 92.60± 2.47% | - |
| Normalizer 1 | Dataset 1 | RBCs Only | 1,748 | 0 | NA | 440,496 | 1, 2, 3, 4 | ✗ | ✗ | ✗ | ✓ | 88.96± 0.35% | 93.84± 1.05% | - |
| Normalizer 2 | Dataset 2 | WBCs Only | 0 | 230 | NA | 74,520 | 1, 2, 3, 4 | ✗ | ✗ | ✗ | ✓ | 81.99± 0.41% | 88.77± 0.15% | - |
| Normalizer 3 | Dataset 3 | Dataset 1 + Dataset 2 | 1,748 | 230 | 7.6 : 1 | 515,016 | 1, 2, 3, 4 | ✗ | ✗ | ✗ | ✓ | 89.94± 0.65% | 94.23± 0.06% | - |
| Normalizer 4 | Dataset 4 | Dataset 3 with image resizing | 1,748 | 230 | 7.6 : 1 | 515,016 | 1, 2, 3, 4 | ✗ | ✗ | ✓ | ✓ | 86.98± 1.26% | 93.10± 0.34% | - |
| Normalizer 5 | Dataset 5 | WBCs Only from high quality images | 0 | 300 | NA | 100,800 | 1, 2 | ✓ | ✗ | ✗ | ✓ | 88.33± 1.32% | 92.72± 0.16% | - |
| Normalizer 6 | Dataset 6 | Dataset 3 + Dataset 5 with different zoom factors | 1,748 | 230 + 300 | 3.3 : 1 | 615,816 | 1, 2, 3, 4 | ✓ | ✓ | ✗ | ✓ | 88.96± 1.22% | 92.64± 0.51% | Better than N1,2,3,4,5 |
| Normalizer 7 | Dataset 7 | RBCs & WBCs from high quality images | 1,000 | 360 | 2.8 : 1 | 456,960 | 1, 2 | ✓ | ✗ | ✗ | ✓ | 90.34± 0.58% | 93.66± 0.37% | The Best |
| Normalizer 8 | Dataset 8 | Dataset 3 + Dataset 7 | 1,748 + 1,000 | 230 + 360 | 4.6 : 1 | 971,976 | 1, 2, 3, 4 | ✓ | ✗ | ✗ | ✓ | 88.57± 1.09% | 93.48± 0.11% | - |
| Normalizer 9 | Dataset 9 | Dataset 7 + More WBCs & RBCs | 1,100 | 460 | 2.4 : 1 | 533,232 | 1, 2 | ✓ | ✗ | ✗ | ✓ | **90.45± 0.34%** | **94.48± 0.37%** | The Best |

Normalizer 1 was trained on Dataset 1 which consists of a collection of 80×80×3 cropped images with an RBC only in the central region of each image. The evaluation of this normalizer showed notable improvements in the RBC classification accuracy over the unnormalized case. However, the visual examination showed poor results because of the reddish discoloration of WBCs and platelets samples and their confusion with the RBCs color patterns. Thus, Dataset 2 was developed using the WBCs only which contain different blue color grades to compensate for this obstacle.

For Normalizer 2, we used Dataset 2 for training with a WBC at the center of each cropped image. The RBC classification accuracy degraded significantly compared to the accuracy obtained without normalization. Also, the visual evaluation revealed some RBCs bluish discoloration patterns that could lead to confusion with platelets. The results of these two normalizers motivated for the construction of a mixed dataset with centrally cropped images of both RBCs and WBCs.

For Normalizer 3, we employed Dataset 3, which combines Dataset 1 and Dataset 2 together to give the normalizer the ability to learn how to restore the correct stain colors of both RBCs and WBCs. The results showed clear enhancements over the aforementioned normalizers. However, the visual analysis uncovered several cases of discoloration which were mainly explained due to the normalizer dependence on the cells size to decide the transformed colors.

For Normalizer 4, data augmentation was applied to Dataset 3 to create Dataset 4 by resizing each training image by a factor within the range of 0.5X to 1.5X to let the normalizer exclude the size factor. Unexpectedly, this normalizer showed poor performance compared to Normalizer 3 in both the visual examination and the classification accuracy. All of the previous datasets were built using images acquired from all of the four employed scanners with remarkable variations in image quality.

For Normalizer 5, we used Dataset 5 which contains centrally cropped WBC samples only obtained from the high-quality Scanners 1 and 2. No samples in this dataset were obtained from the 100K-RBC-pathOlOgics dataset or their corresponding WSIs. Also, in comparison to the previously mentioned datasets, Dataset 5 employed a smaller magnification level that allowed the normalizer to jointly learn the stain patterns of a cell and its neighbors. The visual examination revealed clear improvement compared to the previous normalization models. Also, the RBC classification accuracy was comparably better than that of Normalizer 2.

Thus, creating datasets by using WBCs only from high-quality images led to better normalization performance when compared with using WBCs only but from both low- and high-quality images.

For Normalizer 6, in order to test the effect of using a dataset with mixed zooming levels, we conducted the training on Dataset 6, which combines Dataset 3 and Dataset 5. Dataset 6 has large variations in the magnification level not only because of the magnification differences between the two datasets but also due to the additional random downsizing of the images of the two datasets by factors of 0.4, 0.6, and 0.8. This normalizer showed better results in visual examination compared to all previous models. However, Normalizer 6 showed a slight drop in classification accuracy compared to Normalizer 3.

Results of Normalizer 4 and Normalizer 6 work against the solution of using datasets with different zooming/resizing levels in order to overcome the discovered size-related discoloration associated with this technology. The remaining solution was to provide images covering wide range of possible sizes of RBCs, WBCs, and platelets (from very small to very large) but under the same magnification level.

In Normalizer 7, we used a dataset of high-quality RBCs and WBCs images with no samples obtained from the 100K-RBC-pathOlOgics dataset or their corresponding WSIs. This dataset was enriched by images of different sizes but with no resizing performed. The results showed clear improvement in both the visual examination and the classification accuracy.

For Normalizer 8, we created Dataset 8, as the union of Dataset 3 and Dataset 7. The slight drop in accuracy compared to Normalizer 3 is essentially because of the inclusion of the Dataset 3 training samples, which were of both the low- and high-quality types.

Finally, Normalizer 9 was trained on Dataset 9, which added more RBCs, WBCs, and artifact images to Dataset 7 from the same sources. The visual and quantitative results showed the superiority of this model over all other models. Thus, the experiments for normalization development were concluded.

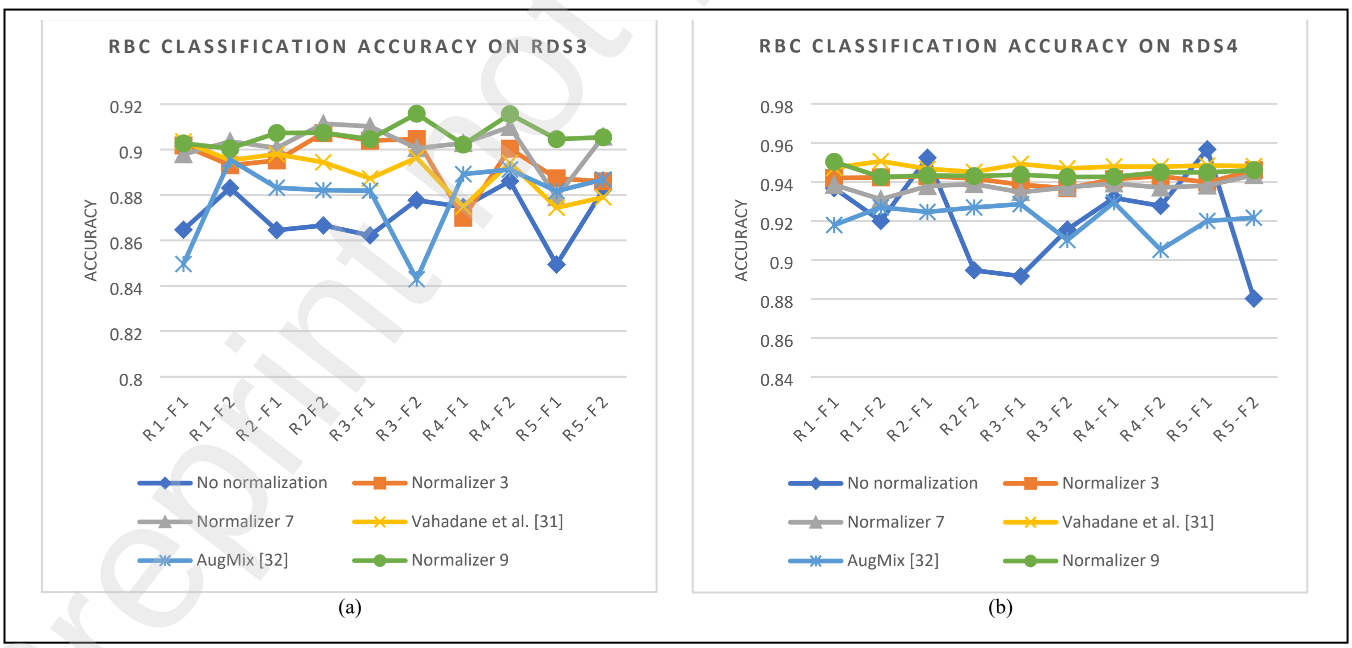



Figure 4. Comparison between RBCs classification accuracies on (a) the RDS3 and (b) the RDS4 datasets using the top three normalization models, the unnormalized classifier, and the two reference methods (Vahadane *et al.* [31] and AugMix [32]). A 5×2 cross-validation scheme is used for comparison. The

*x*-axis represents the round-and-fold combination and the *y*-axis represents the classification accuracy.

TABLE III. AVERAGE PERFORMANCE EVALUATION RESULTS ON THE RDS3 AND RDS4 DATASETS WITH THE TOP THREE NORMALIZATION MODELS AND TWO STATE-OF-THE-ART METHODS. THE BOLD VALUES INDICATE THE LARGEST VALUES IN THE COLUMN.

| **Methods** | **Recall (%)** | | **Precision (%)** | | **F1-score (%)** | | **Accuracy (%)** | | **Worst accuracy (%)** | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **RDS3** | **RDS4** | **RDS3** | **RDS4** | **RDS3** | **RDS4** | **RDS3** | **RDS4** | **RDS3** | **RDS4** |
| **No normalization** | **81.48** | 86.32 | 84.13 | 88.35 | 80.43 | 85.94 | 87.11 | 92.08 | 84.94 | 88.02 |
| **Vahadane** *et al.* [31] | 80.72 | 91.59 | 84.41 | **89.82** | 81.67 | 90.59 | 88.97 | **94.78** | 87.44 | **94.51** |
| **AugMix** [32] | 79.42 | 86.32 | 84.09 | 88.35 | 79.21 | 85.94 | 87.84 | 92.12 | 84.29 | 90.52 |
| **Normalizer 3** | 79.99 | 91.52 | 84.68 | 89.69 | 81.24 | 90.50 | 89.50 | 94.14 | 87.00 | 93.67 |
| **Normalizer 7** | 80.68 | 91.70 | 85.19 | 88.87 | 82.06 | 90.08 | 90.22 | 93.76 | 87.90 | 93.11 |
| **Normalizer 9** | 81.39 | **92.12** | **85.76** | 89.71 | **82.76** | **90.81** | **90.66** | 94.44 | **90.05** | 94.25 |

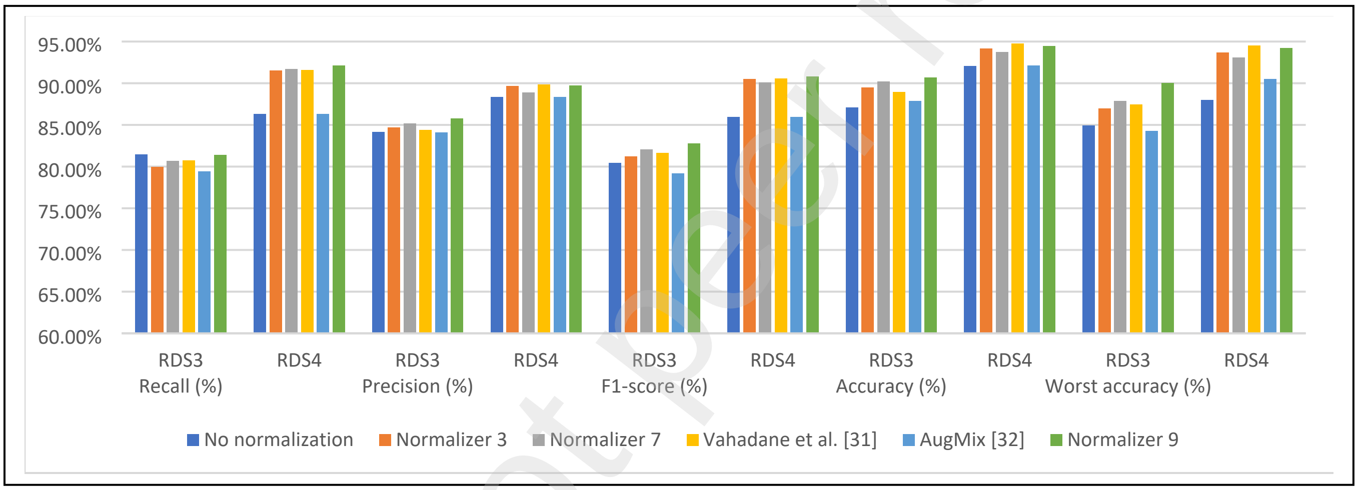



Figure 5. Average performance evaluation results on the RDS3 and RDS4 datasets with the top three normalization models and two state-of-the-art methods.

TABLE IV. STATISTICAL SIGNIFICANCE TEST RESULTS FOR NORMALIZER 9 AGAINST OTHER METHODS IN RBC CLASSIFICATION ON THE RDS3 AND RDS4 DATASETS. THE P-VALUES IN SOLID UNDERLINES INDICATE SIGNIFICANT IMPROVEMENTS OF NORMALIZER 9 AGAINST OTHER METHODS. THE P-VALUE IN DOTTED UNDERLINES INDICATES A SIGNIFICANT IMPROVEMENT IN VAHADANE'S METHOD AGAINST NORMALIZER 9. THE OTHER P-VALUES INDICATE INSIGNIFICANT DIFFERENCES.

| **Methods** | **Recall (%)** | | **Precision (%)** | | **Accuracy (%)** | | **F1-score (%)** | |
|---|---|---|---|---|---|---|---|---|
| | **RDS3** | **RDS4** | **RDS3** | **RDS4** | **RDS3** | **RDS4** | **RDS3** | **RDS4** |
| **No normalization** | 9.2E-01 | **2.3E-09** | **6.5E-04** | **8.2E-04** | **5.9E-08** | **9.6E-03** | **2.0E-02** | **4.1E-08** |
| **Vahadane** *et al.* [31] | 2.9E-01 | **8.2E-03** | **5.1E-04** | 6.3E-01 | **2.2E-04** | **1.2E-03** | **4.6E-02** | 1.9E-01 |
| **AugMix** [32] | 5.9E-02 | **2.3E-09** | **3.1E-06** | **8.2E-04** | **1.3E-04** | **7.4E-08** | **4.7E-04** | **4.1E-08** |
| **Normalizer 3** | 7.1E-02 | **1.7E-03** | **2.3E-03** | 9.6E-01 | **8.9E-03** | **1.7E-02** | **2.8E-02** | 8.9E-02 |
| **Normalizer 7** | 2.8E-01 | **2.0E-02** | 8.2E-02 | **3.9E-03** | 2.1E-01 | **4.3E-05** | 2.2E-01 | **2.4E-03** |

### C. *Effect of normalization on the RBCs classification performance*

#### 1) *Evaluation on data from low-quality scanners*

A 2×2 cross-validation scheme was carried out to evaluate all the convolutional autoencoder models for stain and scan normalization with the RDS3 and RDS4 test sets which are of low quality in staining and scanning, respectively. The average classification accuracy results were reported in Table II. Then, for the top three normalization models (Normalizers 3, 7, and 9), we used a 5×2 cross-validation scheme on RDS3 and RDS4 test sets. RBC classification accuracies for all folds were presented in Fig. 4 and the average performance measures used for evaluation were recall, precision, F1-score, accuracy, and worst accuracy (See Table III and Fig. 5). After that, we tested the statistical significance of the performance results using the two-sample t-test for the difference between the mean performance measures. Tests were carried out for Normalizer 9 against each of all other methods (Reported in Table IV). The results were also compared against two state-of-the-art methods. The first approach, proposed by Vahadane *et al.* [31] used color normalization and sparse stain separation operations. In contrast, the other approach introduced by Google Research at the ICLR2020 conference, used the AugMix augmentation policy [32].

The results revealed that classification models tested on the RDS3 dataset without normalization had an average accuracy of 87.11%. Models tested on RDS3 images after normalization with Normalizer 9 achieved an average accuracy of 90.66% (See Table III and Fig. 5). The classification models tested on RDS4 without normalization had an average accuracy of 92.08%, while models tested on RDS4 images after normalization with Normalizer 9 achieved an average accuracy of 94.44% (See Table III and Fig. 5). Moreover, Normalizer 9 showed superiority among all other methods using the F1-score. The statistical significance testing results with the two-sample t-test are shown in Table IV. Clearly, there were significant differences between the F1-scores obtained on

RDS3 using Normalizer 9 compared to the no-normalization, AugMix, and Vahadane's methods. Also, there were significant differences between the F1-scores obtained on RDS4 using Normalizer 9 compared to the no-normalization and AugMix methods. However, Vahadane's method showed a significant difference in classification accuracy obtained on RDS4 compared to Normalizer 9. We observe insignificant differences in the F1-score obtained on RDS4 between the two methods.

TABLE V. RBCS CLASSIFICATION ACCURACIES (%) ON DATA WITH DIFFERENT SIMULATED/SYNTHETIC DISTORTION PATTERNS. THE BEST PROPOSED NORMALIZER IS COMPARED AGAINST THE NO NORMALIZATION CASE AND TWO STATE-OF-THE-ART METHODS. THE BOLD VALUES INDICATE THE LARGEST VALUES IN THE RAW.

| Distortion | Methods | | | |
|---|---|---|---|---|
| | **No normalization** | **Vahadane *et al.*** [31] | **AugMix** [32] | **Normalizer 9** |
| **No distortion** | **96.07** | 95.12 | 88.58 | 96.00 |
| **Color variations** | 58.26 | 92.24 | 64.47 | **95.56** |
| **Overexposure** | 86.42 | 92.00 | 78.86 | **94.92** |
| **Underexposure** | 53.96 | 94.52 | 78.28 | **94.90** |
| **Defocus** | 92.88 | 90.38 | 80.22 | **94.53** |
| **Average** | 77.51 | 92.85 | 78.08 | **95.18** |

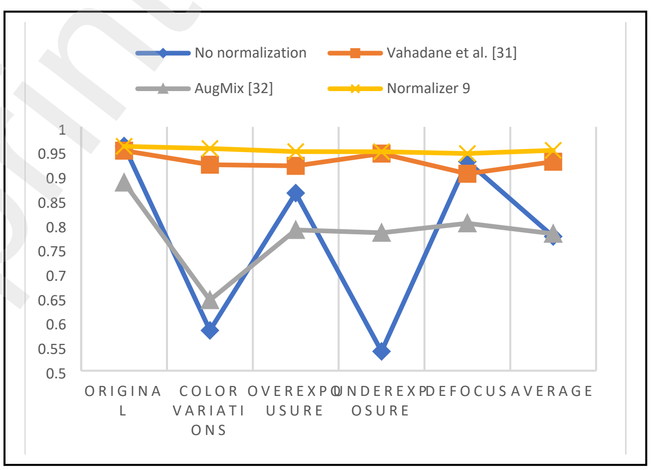


*Figure 6.* Comparison between RBCs classification accuracies on an independent dataset with different distortion types using the top normalization model (Normalizer 9), the no normalization method, and two reference methods (Vahadane *et al.* [31] and AugMix [32]).

*2) Evaluation on data with simulated image distortion patterns*

For further evaluation of the normalization effect, we carried out more RBC classification experiments on an alternate test set that contains 76K images (obtained from the four scanners) under different types of distortion (color variations, overexposure, underexposure, and defocus).

For this simulated/synthetic data, we compared the RBCs classification performance under different distortion types using Normalizer 9 (the best among the proposed autoencoder normalization models), the stain normalization method of Vahadane *et al.* [31], and the AugMix augmentation policy [32]. The results presented in Table V and Fig. 6. reveal that RBC classification without normalization led to an accuracy of only 77.51% (averaged all distortion types). Also, the AugMix method showed low performance with an average accuracy of 78.08%. However, Vahadane *et al.*'s method showed better results with an average accuracy of 92.85%. Still, our Normalizer 9 attained the best average accuracy of 95.18%, with the top performance over all types of distortion. Moreover, it can be seen from the results that color variations and underexposure distortions are the most challenging distortions for the no normalization method and the augmentation-based method. However, normalization-based methods showed considerable improvements in classification performance.

## V. DISCUSSION

In this section, we analyze the implications of using stain and scan normalization in a DL-based RBC classification system according to the obtained results. In this paper, we first explored the effects of the WSI scanning quality on the RBC classification accuracy. In particular, the results in Table I showed clear

performance differences with images obtained from different scanners. On the one hand, Model 1 and Model 2 showed better performance when tested by RDS1 and RDS2 images due to the similarity of their imaging specifications and color staining patterns. So, a classification model trained on RDS1 samples and tested on RDS2 samples (or vice versa) gives generally a high classification accuracy. On the other hand, the color staining patterns of the RDS3 images were visibly different from those in the images of the other scanners. This is why a reduced test accuracy is obtained for a classifier trained without RDS3 samples but tested on such samples. Likewise, the RDS4 images showed unique blurring patterns and color variations that led to the drop in RBC classification accuracy on such images when the classifier had no training samples from RDS4. These results support the assumption that a well-designed preprocessing module is needed in order to normalize all images and to build a robust RBC classification system.
Moreover, different normalization models were trained using simulated/synthetic datasets and validated qualitatively by visually reviewing all the normalized images to check for color discoloration of the RBCs (red) and the WBCs or platelets (blue) samples after normalization. Also, the RBC classification performance with and without normalization was quantitatively evaluated based on the accuracy metric. The results showed the superiority of the Normalizer 9 over the other normalizers in both the qualitative and quantitative evaluations. This normalizer was trained on simulated/synthetic datasets created from best-quality real images containing a wide range of size variations of cells in the RBCs, WBCs, and platelets categories with an RBCs-to-WBCs ratio of 2.4:1. This composition of the training data covered a wide range of color and size variations, and this helped to boost the normalization performance.

As summarized in Table II, different simulated datasets were constructed and used to explore the influence of several factors in building deep-learning-based blood cell image normalizers. The results of Table II led us to the conclusion that having images of essentially the same magnification level will give better normalization and classification results compared to the case when the normalizer is trained on images of widely varying magnification levels.

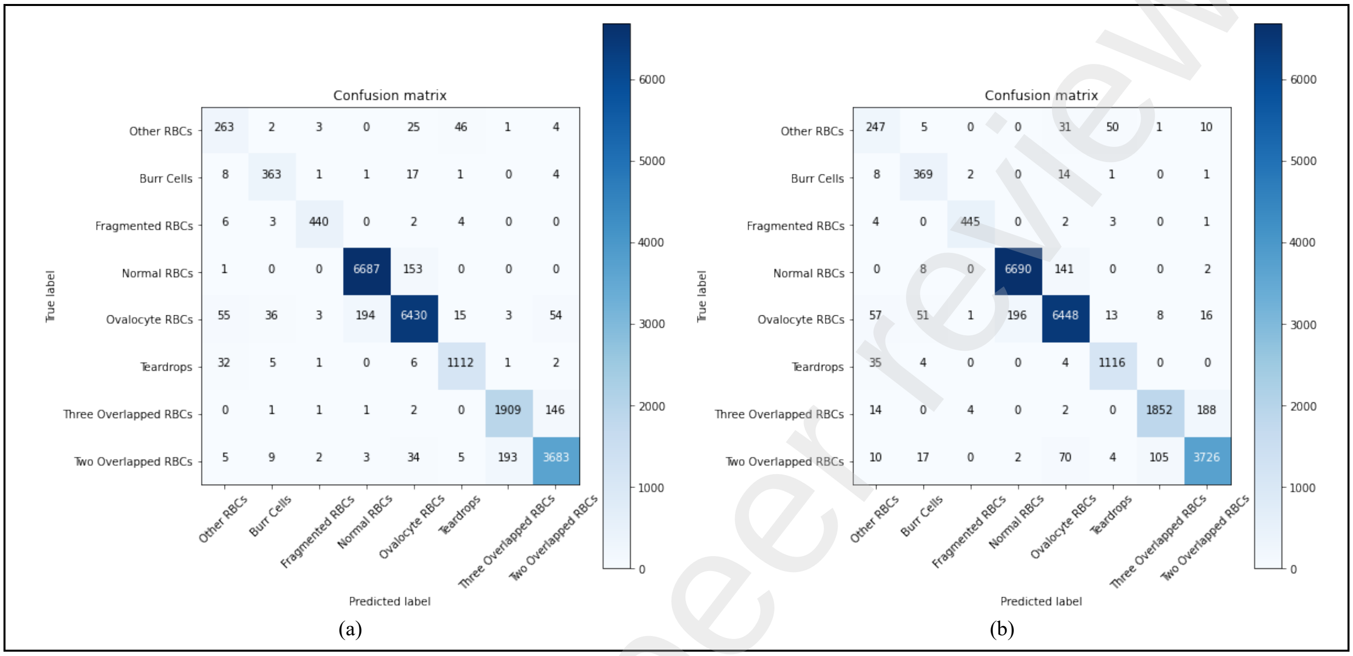



Figure 7. Confusion matrix for best classification accuracy (obtained from RDS4 test set) using: (a) Normalizer 9 achieved an accuracy of 95.03%, (b) Vahadane *et al.* [31] achieved an accuracy of 95.06%.

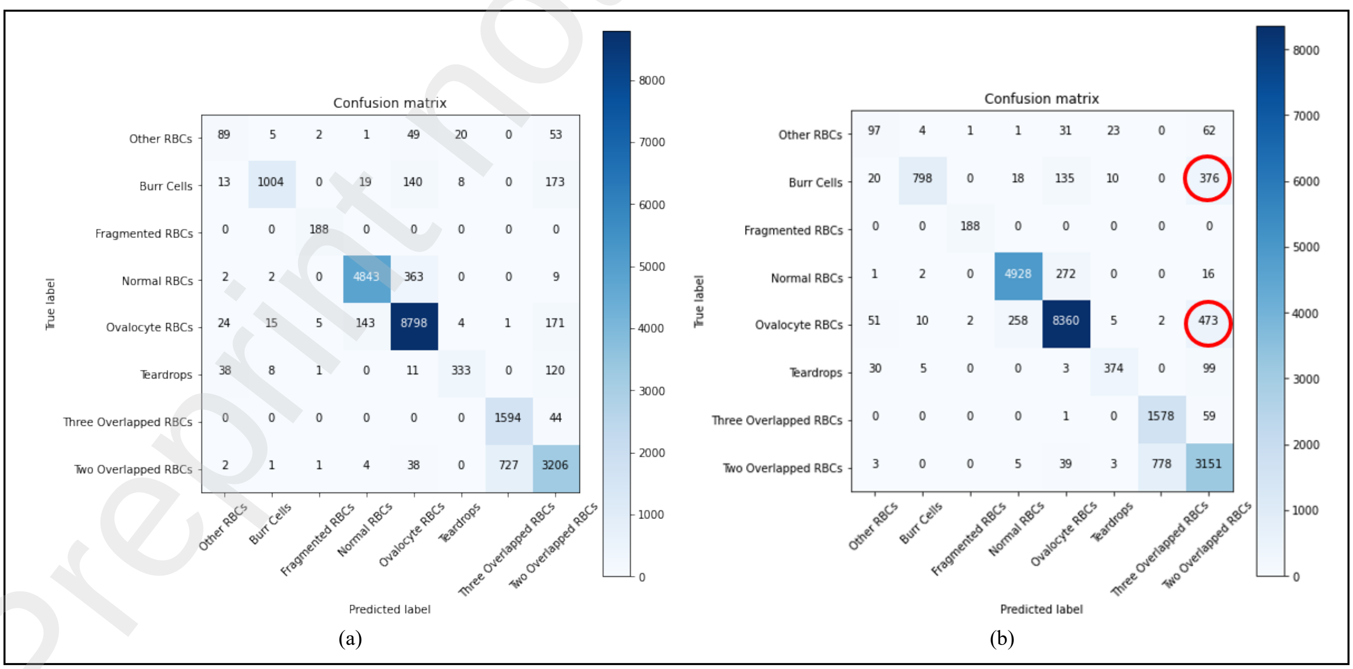


Figure 8. Confusion matrix for worst classification accuracy (obtained from RDS3 test set) using: (a) Normalizer 9 achieved an accuracy of 90.05%, (b) Vahadane *et al.* [31] achieved an accuracy of 87.44%.

Furthermore, using images with a small magnification level where more cells per image were included during the normalizer training was better than using highly magnified ones of few cells per image. Indeed, a normalization model can learn better from inter-cellular variations within low-magnification images. In other words, building a stain normalization model using large patch images with hundreds of cells per image can lead to better results. Also, building training datasets from high-quality images is essential to achieve high performance on stain normalization and subsequently RBC classification (See the results for Normalizer 7 in Table II). Using images collected from both low-quality and high-quality scanners can cause a drop in the performance as seen with Normalizer 8 (in Table II), this normalizer augmented the high-quality Dataset 7 with mixed-quality samples from Dataset 3 which contained low-quality images.

In addition, a balance between the RBCs and WBCs image samples in the normalizer training data is crucial to reach the best normalization results without overlap between the RBCs and WBCs color patterns. For example, in comparison to Dataset 7, Dataset 9 has a more balanced composition of RBCs and WBCs images. Also, Dataset 9 is a rich dataset that encompasses all RBCs and WBCs types besides platelets and artifacts with their natural size variations. These aspects justify the superior robust normalization performance of Normalizer 9 with minimal color overlap and artifacts. Taking all the aforementioned aspects in the construction of training datasets for stain and scan normalization, This shall help improve the performance of blood cell image classifiers on images obtained from scanners of different characteristics and imaging specifications.

Throughout the above experiments, we made comparisons with two state-of-the-art methods: the stain normalization method of Vahadane *et al.* [31], and the

AugMix augmentation policy [32]. The AugMix method was particularly developed by Google Research to improve robustness and uncertainty measures in challenging benchmark image classification problems and also to help models resist unforeseen distortions.

The 5×2 cross-validation results showed high classification performance with Normalizer 9 on both the RDS3 and RDS4 datasets (Table IV). This normalizer can effectively map the color distribution of the images obtained from different scanners to the color distribution of the training images. Also, this normalizer appeared to have the ability to standardize the brightness and contrast variations, and sharpen the blurred images. Vahadane *et al.*'s method had good color normalization and brightness standardization capabilities. Still, while the performance of this method roughly matches ours on the RDS4 dataset, our method is significantly better on the RDS3 dataset (See Table IV).

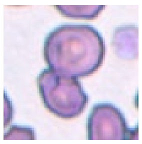
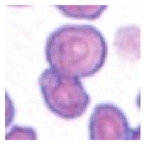
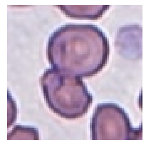
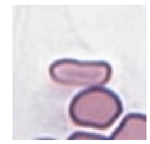
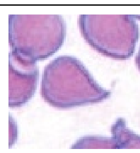
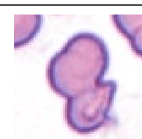
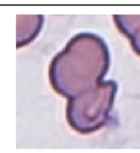
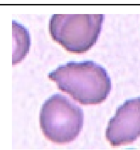
Moreover, we presented confusion matrix for Normalizer 9 and Vahadane *et al.* [31] using best and worst classification accuracy (See Fig. 7 and Fig. 8). In the best classification accuracy, both models showed comparable results with accuracy of 95.03% for Normalizer 9 and 95.06% for Vahadane *et al.* [31]. However, in the worst classification accuracy, Normalizer 9 showed superiority with accuracy of 90.05% compared to 87.44% for Vahadane *et al.* method. The comparison between the confusion matrices (See Fig. 8) showed the ability of our model (Normalizer 9) to correctly identify ovalocytes and burr cells with a high difference of 438 and 206 cells, respectively. Most of these cells are misclassified after using Vahadane *et al.* method as two overlapped cells. The high similarity between the normal RBCs and ovalocytes as well as between the two-overlapped and three-overlapped cells were the main challenges to Normalizer 9.

The reason for using a separate 76K image dataset is to test the performance of the classification models on data with specific more controlled distortion patterns.

The results showed that our best normalization method (Normalizer 9) is clearly superior to Vahadane *et al.*'s method and the AugMix method. However, Vahadane *et al.*'s method showed comparable results for the underexposure case due to brightness adjustment, but a significant drop in performance is observed in the cases of color variations, overexposure, and defocus distortions. The AugMix method achieved an average accuracy comparable to classification without normalization. Also, the AugMix method showed some resistance to underexposure and color variations. Still, this method performed worse than the classifier with no normalization for the cases of defocus, overexposure, and undistorted images.

In Table VI, samples of RBC classification results for no-normalization, Vahadane's method, and Normalizer 9 classifiers are presented. Also, the normalized and segmented images for both RDS3 and RDS4 datasets are presented. The results show the ability of Normalizer 9 to generate images that are consistently normalized in terms of color, brightness, and contrast for both test set scanners. This will help the classifier to identify images whose characteristics are similar to those of the training data. But Vahadane's method classifier and the no-normalization classifier may face test images acquired by cameras whose specifications are different from those of the training data. While the presented samples demonstrated precise segmentation results, using challenging images with OOD characteristics may cause the segmenter's failure in detecting accurate cell contours, and hence the classification error shall increase. This is why the Normalizer 9 classifier is more superior in predicting more challenging RBC images compared to the Vahadane's method classifier and the no-normalization classifier.

TABLE VI. Samples of real RBC images normalized using Normalizer 9 and Vahadane's method. The two methods are compared with the no-normalization classifier. All classifiers are tested on the unseen datasets (RDS3 and RDS4).

| **Dataset** | **RDS3** | | | **RDS4** | | |
|---|---|---|---|---|---|---|
| **Method** | **No normalization** | **Vahadane *et al.* [31]** | **Normalizer 9** | **No normalization** | **Vahadane *et al.* [31]** | **Normalizer 9** |
| **Normalized image** |  |  |  |  |  |  |
| **Segmented image** |  |  |  |  |  |  |
| **True label** | Two overlapped RBCs | Two overlapped RBCs | Two overlapped RBCs | Ovalocyte RBCs | Ovalocyte RBCs | Ovalocyte RBCs |
| **Predicted label** | Three overlapped RBCs | Two overlapped RBCs | Two overlapped RBCs | Angled cells | Burr cells | Ovalocyte RBCs |
| **Normalized image** |  |  |  |  |  |  |
| **Segmented image** |  |  |  |  |  |  |
| **True label** | Angled cells | Angled cells | Angled cells | Three overlapped RBCs | Three overlapped RBCs | Three overlapped RBCs |
| **Predicted label** | Teardrops | Teardrops | Angled cells | Two overlapped RBCs | Two overlapped RBCs | Three overlapped RBCs |
| **Normalized image** |  |  |  |  |  |  |
| **Segmented image** |  |  |  |  |  |  |
| **True label** | Burr cells | Burr cells | Burr cells | Teardrops | Teardrops | Teardrops |
| **Predicted label** | Two overlapped RBCs | Two overlapped RBCs | Burr cells | Burr cells | Burr cells | Teardrops |

In Fig. 9, visual examination samples of normalization errors using Normalizer 9 and Vahadane's method are presented with their corresponding interpretation. The evaluation results reveal that Normalizer 9 is more reliable for visual examination when compared with Vahadane's method. In almost all cases, Vahadane's method failed to preserve the bluish color of the WBC nucleus and the platelets. However, normalization effect on WBC and platelets classification is not investigated in this work. we believe that the identification will not rely only on the absolute colors, but also on the shape, structure, and texture features of the cells. Moreover, simulated images with distortions and added perturbations during classifier training might enable it to correctly classify the WBCs and platelets even after color manipulation.

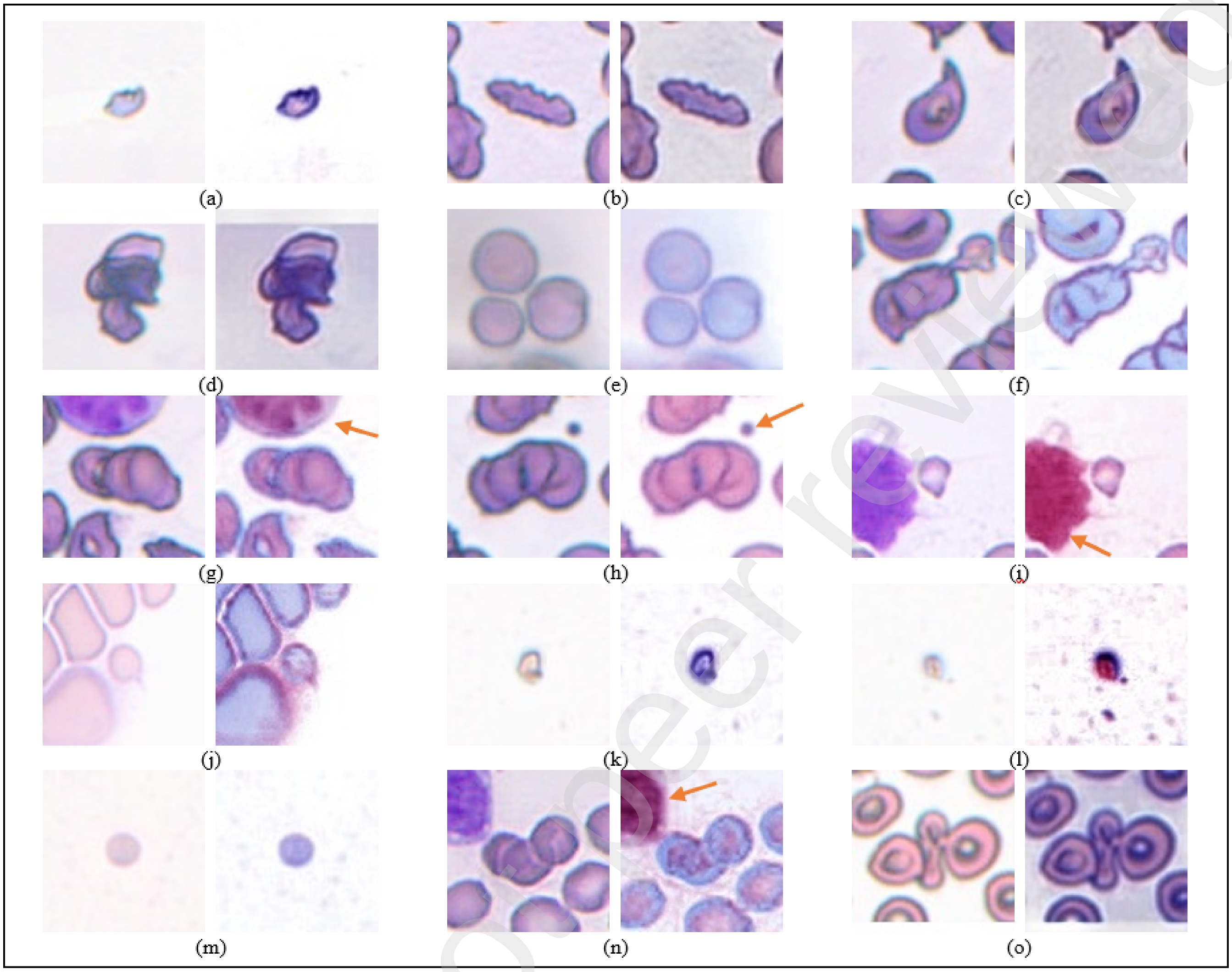



Figure 9. Samples of normalization errors in Normalizer 9 and Vahadane's method. each pair of images shows the original un-normalized image on the left side and the normalized version using either Normalizer 9 or Vahadane's method on the right side. (a) Fragmented RBC from RDS3 normalized by Vahadane shows bluish discoloration of part of a fragmented RBC, (b) Burr cell from RDS3 normalized by Normalizer 9 shows slight bluish discoloration of all of a burr RBC, (c) Teardrop from RDS3 normalized by Normalizer 9 shows slight bluish discoloration of all the cell, (d) Three-overlapped cells from RDS3 normalized by Normalizer 9 show slight bluish discoloration of all the cells, (e,f) Three-overlapped cells from RDS3 normalized by Vahadane present slight bluish discoloration of all the cells, (g) Three-overlapped cells from RDS3 normalized by Vahadane show full red discoloration of adjacent WBC nucleus, (h) Three-overlapped cells from RDS3 normalized by Vahadane show full red discoloration of adjacent platelet, (i) Fragmented RBC from RDS4 normalized by Vahadane shows full red discoloration of WBC nucleus. (j) Fragmented RBC from RDS4 normalized by Vahadane shows slight bluish discoloration of the cells, (k) Fragmented RBC from RDS4 normalized by Vahadane shows bluish discoloration of part of a fragmented RBC, (l) Fragmented RBC from RDS4 normalized by Vahadane shows bluish and red discoloration of part of a fragmented RBC, (m) Fragmented RBC from RDS4 normalized by Vahadane shows full Bluish discoloration of the cell, (n) Three-overlapped cells from RDS4 normalized by Vahadane show full red discoloration of adjacent WBC nucleus, and (o) Three-overlapped cells from RDS4 normalized by Normalizer 9 show slight bluish discoloration of all the cells. In almost all cases, Vahadane's method failed to preserve the bluish color of the WBC nuclei and the platelets.

## VI. Conclusions and Future Directions

### A. General conclusions

Variations of staining and scanning in WSIs can severely affect any RBCs classification system and these variations should be diminished by building a robust image normalization system. In this study, we analyze the impact of stain and scan normalization on an RBC image classification system that utilizes three consecutive DL models. These models are used for stain and scan normalization, cell segmentation, and cell classification, respectively. The models include a convolutional autoencoder, a U-Net architecture, and an EfficientNetB0 architecture. To the best of our knowledge, this is the first effort to address the domain generalization problem in RBC classification using a large and diverse dataset from manually prepared blood smear images.

Collecting real cell datasets that cover all imaging variations can be challenging and impractical. Therefore, it was essential to use simulated or synthetic data with a wide range of size, color, brightness, and blurring variations to create a robust normalization model. The target images for building such a cell normalization model should preferably be collected from high-quality scanners, with all cell types including RBCs, WBCs, platelets, and even artifacts to cover a large range of cell variations. Our experiments show that Normalizer 9 outperforms two state-of-the-art methods on real and simulated RBC datasets for RBC classification (see Tables III and V).

### B. Limitations

Our work still has some limitations. First of all, the 100K-RBC-pathOlOgics dataset was collected from WSIs of only a 40X magnification level. Indeed, the proposed system can show a drop in performance if tested on images of other magnification levels. Also, images of merged cells can be classified as two or three overlapped cells. Theoretically, merged cells require advanced methods to separate the cells and identify each cell type individually, but practically, the hematologists do not do that. Finally, even though cell classification architectures were studied in our previous work [27], we investigated only one DL architecture for image normalization and cell segmentation. A more detailed study shall investigate other alternative architectures.

### C. Future directions

There are several possible extensions to improve this work. First, it would be useful to build a cell detection model that automatically extracts cells from WSIs data and passes the identified images to the RBCs analysis system. Also, an important extension is to build a WBC classification system to extensively and quantitatively evaluate the normalization effect on the WBC classification performance [33], [34], [35], [36], [37]. As mentioned earlier, the merged-cell problem could be a challenging part of any cell analysis system, and this problem needs to be effectively solved [20], [38]. Finally, the employed DL models need to be compared with other state-of-the-art DL architectures.

## DATA AVAILABILITY

The 100K-RBC-PathOlOgics and 100K-RBC-Mask-PathOlOgics datasets used in this work represent an early version of the recently published PathOlOgics_RBCs datasets https://doi.org/10.6084/m9.figshare.24119511.v1.

## DECLARATION OF COMPETING INTEREST

The second and third authors (AE & YA) are cofounders of PathOlOgics, a limited-liability company for processed medical data and tele-pathology services. The other authors have no relevant financial or non-financial interests to disclose.

## FUNDING

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

## AUTHOR CONTRIBUTIONS

All authors contributed to the study conception and design. Material preparation and data collection were performed by AE and YA. Data analysis and the first draft were completed by ME. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

## APPENDIX A. 100K-RBC-PATHOLOGICS: MULTI-SCANNER RBC DATASET [27]

The two hematologists curated large datasets of RBC images, each with its corresponding ground-truth binary semantic segmentation mask. They selected 25 manually prepared and stained peripheral-blood and bone-marrow smears from patients suspected to have primary myelofibrosis (PMF) of the bone marrow. For generating whole slide images (WSIs), four digital pathology scanners were utilized. Each scanner was equipped with a different light microscope using 40X magnification power, control units, integrated cameras, and software for patch stitching. The 25 slides were divided into 4 subsets and each scanner was used to scan a different subset. Table A.1 illustrates the distribution of the 25 slides among the four scanners. The physical dimensions of each slide were about 1×2 inches. Each slide was subdivided into around 2,000 non-overlapping patches with a size of 539×1076 pixels. Each patch had more than 200 blood cells. To create masks, the hematologists developed a custom semi-automatic segmentation scheme as follows. First, the cellular borders are manually outlined using a digital pen. Then, the identified cell within a cropped image is automatically centered and the mask is precisely adjusted to match the central position of the cell.

The first dataset, "100K-RBC-Mask-PathOlOgics," comprised 100,118 cropped RBC images, each paired with its corresponding mask. This dataset was used for developing the automated RBC cell segmentation algorithm. On the other hand, the second dataset, "100K-RBC-PathOlOgics," included 100,873 segmented RBC images and was utilized for classification purposes.

The labeling criteria were designed to focus on clinically significant RBC types, and where visual examination is considered exclusive and unassisted by current technological solutions. After each hematologist independently labeled the complete dataset, a final discussion was conducted to address and resolve any discrepancies in the labeling process. Each cell was categorized into one of eight classes, including normal/rounded RBCs, ovalocytes (oval or egg-shaped), burr cells (crenated), fragmented RBCs, teardrop-shaped RBCs, two-overlapped RBCs, three-overlapped RBCs, and other RBCs that contain artificial/false teardrops. The presence of ovalocytes exceeding 5% of the total RBCs is associated with almost all types of anemia. Additionally, teardrops and fragmented RBCs are linked to serious medical conditions and should never be underestimated, as they could be fatal. Lastly, burr cells indicate a poor staining process if their medical causes are excluded. The utilization of manually prepared smears, which is prevalent in medical labs worldwide, may result in some areas or fields on the WSI being unsuitable for RBC examination and counting. The selection of optimal counting areas depends on evaluating the ratio between individual cells and overlapping cells. Regions with a lower occurrence of overlapping cells are considered more appropriate for accurate examination and counting purposes.

This dataset has several key advantages. First, a well-trained classifier can effectively operate on manually prepared and stained smears without mandating prior standardization of staining or smearing because of the diversity in the dataset. Also, such a classifier can serve as a sensitive screening tool for anemia, based on the percentage of ovalocytes among the total RBCs, while also exhibiting high specificity for identifying teardrops and fragmented RBCs. At the same time, the tool demonstrates sensitivity to detecting improper manual staining, as indicated by the burr cells percentages. Furthermore, the classifier is capable of identifying the most suitable areas to initiate counting, relying on the ratio between overlapping and individual cells.

Samples of the RBC images from the eight classes are shown in Fig. A.1. Figure A.2 shows the distribution of the collected images among the eight RBC classes and Fig. A.3 presents samples of RBC images with their corresponding segmentation masks. Notably, each of the selected 25 smears contained samples from all targeted RBC classes, exhibiting variations in class proportions. As a result, images representing every RBC class were successfully collected from each WSI. Figure A.4 illustrates the appearance and color variations among images collected by the four scanners.

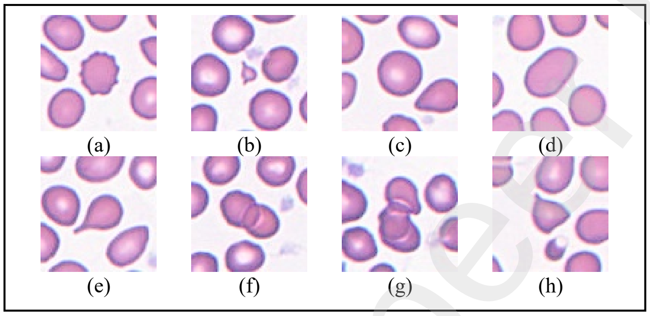



Figure A.1. Samples of eight RBC types. (a) Burr cells, (b) Fragmented RBCs, (c) Normal RBCs, (d) Ovalocytes, (e) Teardrops, (f) Two overlapped RBCs, (g) Three overlapped RBCs, and (h) Other RBC types.

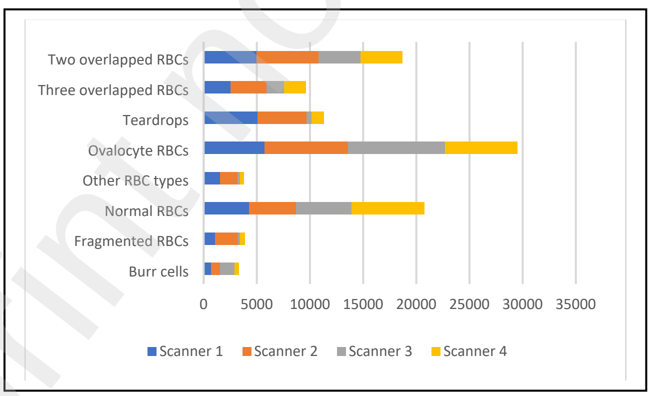



Figure A.2. Distribution of 100K cells among eight RBC classes and four scanner types.

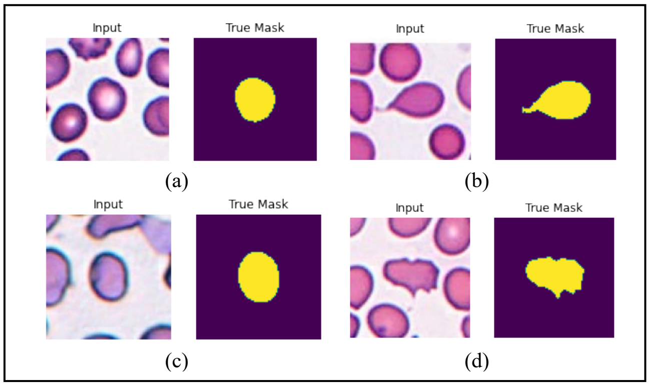



Figure A.3. Samples of RBCs with their corresponding ground-truth masks. (a) Normal RBC, (b) Teardrop, (c) Ovalocyte, and (d) Burr cell.

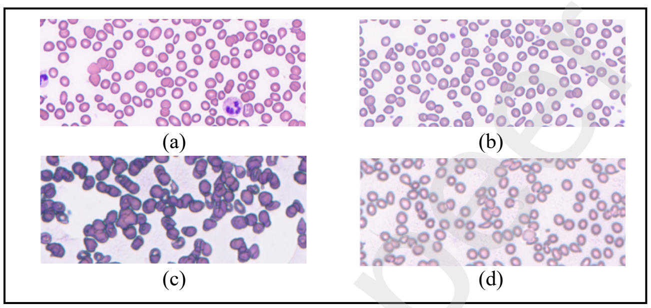



Figure A.4. RBC image patches collected from four different scanners: (a) Images from Scanner 1, (b) Scanner 2, (c) Scanner 3, and (d) Scanner 4.

Table A.1. Distribution of the 25 WSIs among the four scanners.

| **Scanners** | **WSIs** | | | | | | |
|---|---|---|---|---|---|---|---|
| **1** | 1 | 5 | 6 | 8 | 25 | | |
| **2** | 2 | 3 | 4 | 7 | 9 | 11 | |
| **3** | 14 | 15 | 19 | 20 | 22 | 23 | 24 |
| **4** | 10 | 12 | 13 | 16 | 17 | 18 | 21 |

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# Normalization of Stain and Scan Distortions for Red Blood Cell Analysis on Manually Prepared Smears Using Deep Learning

Mohamed Elmanna [a], Ahmed Elsafty [b], Yomna Ahmed [b], Muhammad Rushdi [a, c*], and Ahmed Morsy [a]

[a] Department of Biomedical Engineering and Systems, Faculty of Engineering, Cairo University, Cairo University St, Giza 12613, Egypt

[b] PathOlOgics, LLC for processed medical data and tele-pathology services, Egypt

[c] School of Information Technology, New Giza University, Giza 12256, Egypt

**Abstract**

The introduction of the digital pathology realm has provided numerous opportunities for computer-aided diagnosis (CAD), leveraging advanced machine learning techniques such as deep learning (DL). However, in hematopathology, challenges arise from staining variations and scanner distortions, likely cause CAD performance degradation. In this work, we investigate the effects of stain and scan normalization on red blood cell (RBC) image analysis systems. The normalized images are thus used for DL-based RBC segmentation and classification. The 100K-RBC-PathOlOgics dataset was used for RBC segmentation and classification. Then, large synthetic datasets were created to reflect the wide data variability among scanners, staining protocols, and imaging conditions. Subsequently, an autoencoder model was trained using the simulated data to perform normalization. Our DL-based normalizer significantly improves RBC classification F1-score compared to the classifier without normalization, especially for low-quality images from Scanner 3 (with a *p*-value of $\mathbf{2.0 \times 10^{-2}}$) and Scanner 4 (with a *p*-value of $\mathbf{4.1 \times 10^{-8}}$).

***Keywords:*** Whole slide images, hematopathology, stain normalization, scanner distortion normalization, deep learning, RBC classification.

*** Corresponding Author:**

Muhammad Rushdi, PhD
Associate Professor
Department of Biomedical Engineering and Systems
Faculty of Engineering, Cairo University
Cairo University St, Giza 12613, Egypt
Phone: +201064715798
Email: mrushdi@eng1.cu.edu.eg
Homepage: http://scholar.cu.edu.eg/mrushdi
LinkedIn: https://www.linkedin.com/in/muhammad-rushdi-4817a462/
Google Scholar: https://scholar.google.com/citations?user=vDgGLjsAAAAJ&hl=en&oi=ao
ResearchGate: https://www.researchgate.net/profile/Muhammad_Rushdi